disease-causing microorganism and is often made from weakened or attenuated or killed forms of the microbes, their toxins, or one of its surface proteins. Vaccines **"teach"** our body how to defend itself when viruses or bacteria, invade it. Vaccines deliver only very little amounts of inactivated or weakened viruses or bacteria, or parts of them. This allows the immune system to recognize the organism **without actually experiencing the disease**. Some vaccines need to be given more than once (i.e., a 'booster' vaccination) to make sure the immune system can overcome a real infection in the future.

Vaccine initiates the immunization process. The vaccines are classified as first, second and third generation vaccines.

First generation vaccine is further subdivided into live attenuated vaccine, killed vaccine and toxoids **(Fig. 8.9). Live attenuated vaccines** use the **weakened (attenuated), aged, less virulent** form of the virus. E.g. Measles, Mumps and Rubella (MMR) vaccine and the Varicella (chickenpox) vaccine, **Killed (inactivated) vaccines** are killed or inactivated by heat and other methods. E.g. Salk's polio vaccine. **Toxoid vaccines** contain a **toxin or chemical** secreted by the bacteria or virus. They make us immune to the harmful effects of the infection, instead of to the infection itself. E.g. DPT vaccine (Diphtheria, Pertussis and Tetanus).

Second generation vaccine contains the pure surface antigen of the pathogen. E.g.Hepatitis-B vaccine. **Third generation vaccine** contains the purest and the highest potency vaccines which are synthetic in generation. The latest revolution in vaccine is **DNA vaccine** or **recombinant vaccine** (Refer Chapter- 10 for details).



Vaccino therapy is the method of use of vaccine for treatment of disease. Dr. Edward Jenner prepared first vaccine for small pox in 1796. Polio vaccine was developed by Dr. Jonas Salk (vaccine consists of inactivated microorganism) and Dr. Albert Sabin (live attenuated oral polio vaccine). Louis Pasteur (1885) discovered vaccine against rabies, anthrax and cholera. BCG vaccine was developed by Calmette and Guerin against tuberculosis in France in the year 1908.

8. 10 Vaccination and immunization

"**Vaccination** is the process of administrating a vaccine into the body or the act of introducing a vaccine into the body to produce immunity to a specific disease." **Immunization** is the process of the body building up immunity to a particular disease. Immunization describes the actual changes in the body after receiving a vaccine. Vaccines work by fighting the pathogen and then recording it in their **memory system** to ensure that the next time this pathogen enters the body, it is eliminated far quickly. Once, the body is able to fight against the disease, it is believed to have built the immunity for it, also known as the body being immunized against the disease.

8.11 Hypersensitivity

Some of the individuals are very sensitive to some particles present in the environment. The exaggerated response of the immune system to certain antigens present in the environment is called **allergy (allo-altered, erg-reaction)**. The substances to which such an immune response is produced are called **allergens**. An allergen is an antigen that causes an allergic

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